



Live Production Deployment Available

All features, database models, and doctor workstations are live online.

<https://yencare-platform.vercel.app>

1. PLATFORM ACCESS & TEST ACCOUNTS

Hosted & Local Endpoints

- **Hosted Web:** <https://yencare-platform.vercel.app/>
- **Hosted Staff:** <https://yencare-platform.vercel.app/staff/login>
- **Hosted Board:** <https://yencare-platform.vercel.app/queue>
- **Local Dev:** <http://localhost:5173> (Backend: port 4000)

Operating Hours (Accra Time)

- **Students' Clinic:** Mon–Fri, 08:00–16:00 (Weekends closed)
- **KNUST Hospital:** 24/7 round-the-clock
- **Demo Toggle:** Staff portal includes *Today & Seed Day (2026–09–15)*.

ROLE	STAFF MEMBER	EMAIL / STAFF ID	ROOM	DEFAULT PASSWORD	CORE SCOPE
RECEPTIONIST	Abena Osei	abena.osei@yencare.gh / stf_01	Front Desk	yencare	Roster triage, check-in, walk-in intake
DOCTOR 1	Dr. Kwame Boateng	kwame.boateng@yencare.gh / stf_02	Room 1	yencare	Room 1 queue, call next, complete visit
DOCTOR 2	Dr. Ama Serwaa	ama.serwaa@yencare.gh / stf_04	Room 2	yencare	Room 2 queue, call next, complete visit
ADMIN	Kojo Mensah	kojo.mensah@yencare.gh / stf_03	Admin Desk	yencare	System oversight, clinical policy audit

2. CLINICAL LIFECYCLE WORKFLOW

1. Student Booking

Student picks clinic, doctor, date & time slot. Immediate YC-XXXX booking reference and SMS alert issued.

2. Arrival & Queue

Student arrives; status updates to WAITING. Room-prefixed ticket token (#A1, #B1) is generated.

3. Consultation

Doctor calls patient (CALLED), board directs student to room, doctor completes consultation (COMPLETED).

3. STEP-BY-STEP TESTING JOURNEYS

Journey 1: Student Online Booking Experience

1. Go to `https://yencare-platform.vercel.app/` (or localhost). Click "**Book Consultation**".
 2. **Select Clinic Site:** Choose **KNUST Students' Clinic**.
 3. **Enter Details:** Full Name: `Akosua Mensah`, Phone: `024 123 4567`, Student Index: `20612345` (8 digits).
 4. **Select Service & Doctor:** Choose **General OPD** and select **Dr. Kwame Boateng (Room 1)**.
 5. **Select Date & Slot:** Pick an upcoming weekday (e.g. tomorrow) and an available time slot (e.g., `09:30`).
 6. **Confirm Booking:** Review summary details and click "**Confirm booking**".
- ✓ **Verification:** Confirms with unique Booking Reference (e.g. `YC-4821`). Outbound SMS fires to phone. Reference saved in local browser storage.

Journey 2: Self-Service Reschedule & Cancel (SMS OTP)

1. Navigate to **Appointments** → **Find Existing Appointment**. Search using your Reference Code or Student Index.
 2. **Reschedule Flow:** Click "**Reschedule Appointment**" → "**Send Verification Code**". Enter the 4-digit SMS OTP. Pick a new date/time slot and submit.
 3. **Cancel Flow:** On another booking, click "**Cancel Appointment**". Enter OTP, select cancellation reason, and confirm.
- ✓ **Verification:** Reschedule updates date/time under identical reference code and frees old slot. Cancel sets status to `CANCELLED` and unlocks the slot immediately.

Journey 3: Patient Arrival & Live Queue Token

1. On visit day, open the appointment card and click "**I Have Arrived**" (active 60 mins early to 15 mins late).
2. Status advances: `BOOKED` → `CHECKED_IN` → `WAITING`.
3. A sequential token is assigned (e.g., `#A1` for Room 1, `#B1` for Room 2) with estimated wait time.
4. Open Corridor Display (`/queue`) in a separate tab; verify the token appears under the live queue.

Journey 4: Reception Desk Operations (Staff Portal)

1. Log into `/staff/login` as **Abena Osei** (`abena.osei@yencare.gh` / `yencare`).
2. **Roster Triage:** Filter by `All`, `Booked`, `Checked-In`, `Waiting`, `Walk-In`.
3. **Desk Check-In:** For an in-person arrival, click "**Check In**" → "**Send to Queue**".
4. **Fast-Track Walk-In:** Click "**+ Add Walk-In**". Enter student details, assign clinician, and submit. The walk-in is immediately slotted, checked in, and queued.

Journey 5: Doctor Workstation & Consultation

1. Open an incognito window and sign in as **Dr. Kwame Boateng** (`kwame.boateng@yencare.gh`). Workstation defaults to **Room 1**.
2. **Clinician Isolation (Critical):** Verify Dr. Kwame sees *only* his Room 1 queue. He cannot see or call Dr. Ama's patients.
3. **Call Next:** Click "**Call Next Patient**". Status moves to `CALLED`; Corridor Display directs patient to Room 1.
4. **Complete Visit:** Click "**Complete Visit**". Status moves to `COMPLETED`; room resets for the next ticket.
5. Log in as **Dr. Ama Serwaa** (`ama.serwaa@yencare.gh`) in **Room 2** to confirm separate queue isolation.

4. CRITICAL GUARDRAILS & POLICY STRESS TESTS

Please execute these specific test cases to confirm platform hardening and clinical safeguards:

❑ Student Booking Cap (Max 2 Active Bookings):

Attempt to book 3 active upcoming visits for the same Student Index (20612345).

👉 Expected: 3rd booking rejected with HTTP 409: "You have reached the maximum of 2 active appointments. Please complete or cancel existing visits."

❑ Concurrent Double-Booking Slot Collision:

Two testers open the same doctor and time slot simultaneously and click confirm at the exact same second.

👉 Expected: Exactly one wins (201 Created). The second receives 409 Conflict with friendly "Slot Taken" reassignment.

❑ Clinic Operating Hours Enforcement:

Attempt to book KNUST Students' Clinic on a Saturday or Sunday, or after 16:00.

👉 Expected: Blocked client-side and rejected by server with operating hours notice.

❑ Arrival Window Enforcement:

Try checking in >60 minutes early or >15 minutes late.

👉 Expected: Early arrival directs patient to wait; late arrival flags no-show risk and directs to reception desk.

❑ Invalid OTP Verification:

Enter an invalid code (e.g. 9999) during cancellation or reschedule.

👉 Expected: Rejected with "Invalid or expired verification code". Action blocked until verified.

❑ Simulated Offline Roster Display:

In DevTools → Network tab, switch to **Offline** while on the Staff Roster page.

👉 Expected: Displays "Could not load appointment roster. [Retry]" with a working retry trigger.

5. HOW TO LOG ISSUES DURING TESTING

Bug Report Format

1. **Screen / Component:** (e.g. Doctor Workstation → Call Next)
2. **User Role & Account:** (e.g. Dr. Kwame Boateng, Room 1)
3. **Steps to Reproduce:** (Exact sequence of actions taken)
4. **Expected Result:** (What clinical rule should have occurred)
5. **Actual Result:** (Error message, visual defect, or unexpected status)
6. **Attachments:** (Console error log or screenshot)